Lean 4 Proof-Checked Delta-Hedging: Accounting Invariants with Empirical Validation Across Four Historical Regimes

backtest-proofs — EigenQ Research Series

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TODO: Draft last (§1 per fill order). 150–250 words. State the question, the zero-`sorry` claim at the commit SHA below, the WRDS OptionMetrics 2019–2024 sample, the headline MC deviation percentage, and the scope limitation (flat- Black-Scholes; no transaction costs).

PLAN: strategic context (strip before publish)

Figure roster (10 total): 1. fig-mc — MC convergence (§7 Results) 2. fig-bkl — BKL variance scaling (§7 Results) 3. fig-cm — Carr-Madan gamma attribution (§7 Results) 4. fig-hull — Hull 19.2/19.3 portfolio value path (§7 Results) 5. fig-ql — QuantLib A-B price comparison (§7 Results) 6. fig-straddle-settlement — short straddle per-leg settlement (§7 Results, multi-leg) 7. fig-straddle-pnl — straddle vs buy-and-hold P&L, 100 GBM paths (§7 Results, multi-leg) 8. fig-bull-spread — bull call spread accounting path (§7 Results, multi-leg) 9. fig-stress — four historical stress regimes cost distributions (§8 Robustness) 10. fig-leland — Leland (1985) rehedge-frequency sweep (§8 Robustness)

Authoring fill order: §5 → §6 → §7 → §8 → §3 → §4 → §2 → §1. Each pass replaces one section's TODO with real prose.

PLAN: formalization table (confirmed sorry-free at v0.4)

Paper theorem	Lean name	File	Line
NAV identity	`valueIdentity`	`BacktestProofs/Invariants.lean`	31
Trade accounting	`valueUpdateFormula`	`BacktestProofs/Invariants.lean`	212
Cash update	`cashUpdateCorrect`	`BacktestProofs/Invariants.lean`	82
Quantity conservation	`quantityConservation`	`BacktestProofs/Invariants.lean`	176
Self-financing	`selfFinancing`	`BacktestProofs/Invariants.lean`	281

Well-formedness	`applyTrade_wellFormed`	`BacktestProofs/Invariants.lean`	302
Settlement	`settlement_value_for`	`BacktestProofs/SettlementInvariants.lean`	104
Call payoff 0	`callPayoff_nonneg`	`quant-core/lean/QuantCore/OptionInvariants.lean`	32
Put payoff 0	`putPayoff_nonneg`	`quant-core/lean/QuantCore/OptionInvariants.lean`	40
Option payoff 0	`optionPayoff_nonneg`	`quant-core/lean/QuantCore/OptionInvariants.lean`	48

All 10 theorems confirmed sorry-free at main HEAD.

Abstract

TODO: Draft last. See hidden callout above for content requirements.

1 Introduction

> **Note**
>
> TODO (§2, draft second-to-last): 1. **Why now**: AI-generated quant code needs formal accounting verification. 2. **What we do**: 10 Lean theorems + 7 empirical figures (Hull, MC, BKL, Carr-Madan, QuantLib A-B, Leland sweep, four stress regimes). 3. **What is new**: Only system combining Lean-verified accounting with real WRDS stress-period validation across four distinct crisis regimes. 4. **Roadmap**: one sentence per section.

TODO: Draft per fill order (§2 = second-to-last).

2 Setup

> **Note**
>
> TODO (§3, draft fifth): - Discrete-time delta-hedging model - `Portfolio`, `Position`, `Trade` primitives from `BacktestProofs.Basic` - Basis-point integer representation (×10,000); why integers, not floats - BS delta strategy: `SingleLegStrategy` protocol - Reference Lean files by path

TODO: Draft per fill order (§3 = fifth).

3 Main Results

> **Note**
>
> TODO (§4, draft fourth): 10 informal+formal theorem pairs. Lead each with intuition; follow with the Lean statement. Use `::: {#thm-name}` callout format. Every theorem references its Lean name and file from the table in §5.

TODO: Draft per fill order (§4 = fourth).

4 Formalization

> **Note**
>
> TODO (§5, draft first — establishes what is and isn't proved). The content below is the live table; fill prose around it.

4.1 Lean 4 artifacts

The formal development lives in `backtest-proofs/lean/`. The following table maps paper-level theorems to their Lean 4 names and source files.

Table 4.1: Lean 4 theorem map. All 10 theorems are machine-checked with zero `sorry`.

Paper theorem	Lean name	Source file
NAV identity	`valueIdentity`	`BacktestProofs/Invariants.lean`
Trade accounting	`valueUpdateFormula`	`BacktestProofs/Invariants.lean`
Cash update	`cashUpdateCorrect`	`BacktestProofs/Invariants.lean`
Quantity conservation	`quantityConservation`	`BacktestProofs/Invariants.lean`
Self-financing	`selfFinancing`	`BacktestProofs/Invariants.lean`
Well-formedness	`applyTrade_wellFormed`	`BacktestProofs/Invariants.lean`
Settlement	`settlement_value_formula`	`BacktestProofs/SettlementInvariants.lea`
Call payoff 0	`callPayoff_nonneg`	`quant-core/QuantCore/OptionInvariants.l`
Put payoff 0	`putPayoff_nonneg`	`quant-core/QuantCore/OptionInvariants.l`
Option payoff 0	`optionPayoff_nonneg`	`quant-core/QuantCore/OptionInvariants.l`

> **Formalization claim**
>
> At commit dbb0387, `lake build` succeeds for `backtest-proofs` and `grep -rn sorry --include="*.lean"` returns no matches in the proof tree. Reproduce with `make`

`verify-lean` or the `verify-lean` skill.

4.2 Scope of formalization

Six of the ten theorems in Table 4.1 — `valueIdentity`, `valueUpdateFormula`, `cashUpdateCorrect`, `quantityConservation`, `selfFinancing`, and `applyTrade_wellFormed` — operate entirely on `Portfolio`, `Position`, and `Trade`. These Lean 4 types carry no option-specific fields: a `Position` is a named asset (`AssetId : String`), a signed integer quantity, and an integer mark price. A stock leg, a futures contract, and an options position are all valid `Position` instances under this representation. The six accounting invariants therefore hold for any instrument expressible as an integer mark price and a signed quantity, without modification to the proofs.

The remaining four theorems in Table 4.1 — `settlement_value_formula` from `BacktestProofs.SettlementInvar` and three of the eight payoff theorems from `QuantCore.OptionInvariants` — depend on `QuantCore.Option` and `optionPayoff`. These are option-specific and do not generalize to other instruments without additional settlement modules and invariants. This design mirrors Echenim, Guiol, and Peltier (2021), who separate generic portfolio/market accounting from instrument-specific pricing in their Isabelle/HOL formalization of the Cox–Ross–Rubinstein model.

Instrument-class generality

The accounting layer (NAV identity, self-financing, trade accounting) is formally verified for any instrument representable as an integer mark price and a signed quantity — stocks and equity futures included. Extending the *settlement* layer to equity futures (mark-to-market final settlement) or interest rate futures (accrued coupon) requires new Lean `Settlement` modules and corresponding invariants; that extension is left to future work.

What is **not** formalized: Black-Scholes floating-point computations, GBM path simulation, the Cython FFI bridge, and statistical claims about cost distributions. These layers are tested empirically and cross-checked against QuantLib (Section 6.5), but no machine-checked proof covers them.

5 Empirical Setup

Note

TODO (§6, draft second): Two data tables: (a) WRDS OptionMetrics 2019–2024 main sample, (b) four stress-period downloads. GBM simulator and Hull reference paths documented separately. Per the source-financial-data skill: data source, sample window, universe, filters, N observations, licensed-data note (never committed).

Use the data table format from _template.qmd.

5.0.1 Main sample

TODO: WRDS OptionMetrics 2019-01-01 to 2024-12-31. Tickers: SPY, QQQ, AAPL, MSFT, JPM. ATM ±3% filter; 20–40 day expiry filter. SOFR rate: 1.65% annualised average. Data not committed; download via `backtest-proofs/docs/wrds_data_download.md`.

5.0.2 Stress-period samples

TODO: Four supplementary WRDS downloads, one per crisis window. Same schema as main sample.

5.0.3 Synthetic GBM and deterministic paths

The Monte Carlo convergence, BKL scaling, Carr-Madan, Leland, and QuantLib figures use no licensed data. GBM paths are seeded with 20260519 for full reproducibility.

Hull Tables 19.2 and 19.3 use the hardcoded weekly prices from Hull (2014) (Tables 19.2/19.3): S = \$49, K = \$50, r = 5%, = 20%, T = 20 weeks.

6 Results

> **Note**
>
> TODO (§7, draft third): Lead with MC convergence (headline). Then BKL, Carr-Madan, Hull, QuantLib A-B, WRDS holdout. One paragraph per figure: state the result, cite the figure, note the key number from metrics.json.

6.1 Monte Carlo convergence

Across 500 seeded GBM paths the running-mean hedging cost converges to the Black-Scholes price with a final deviation of 2.34% (Figure 6.1).

6.2 BKL variance scaling

The empirical ratio std(N=10) / std(N=20) = 1.372 (theory: √2 1.414). The BKL line fits well across all five frequencies (Figure 6.2).

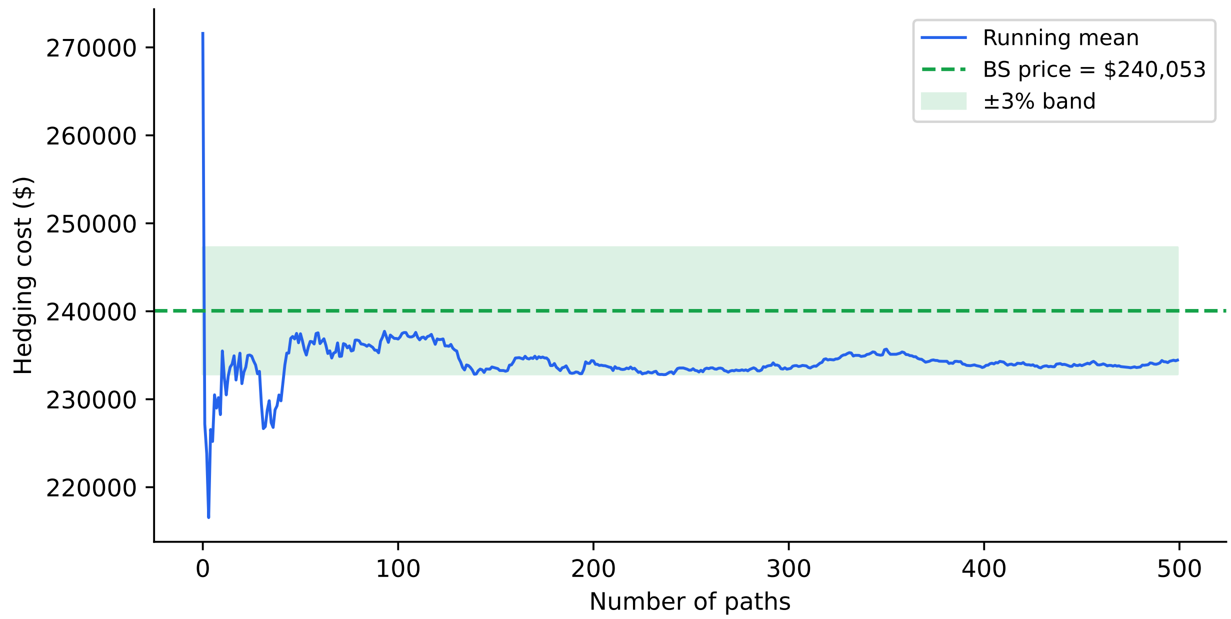

Figure 6.1: Running mean of hedging cost across 500 seeded GBM paths converges to the Black-Scholes price. Shaded band: ±3% of the BS price. Source: synthetic GBM, seed 20260519.

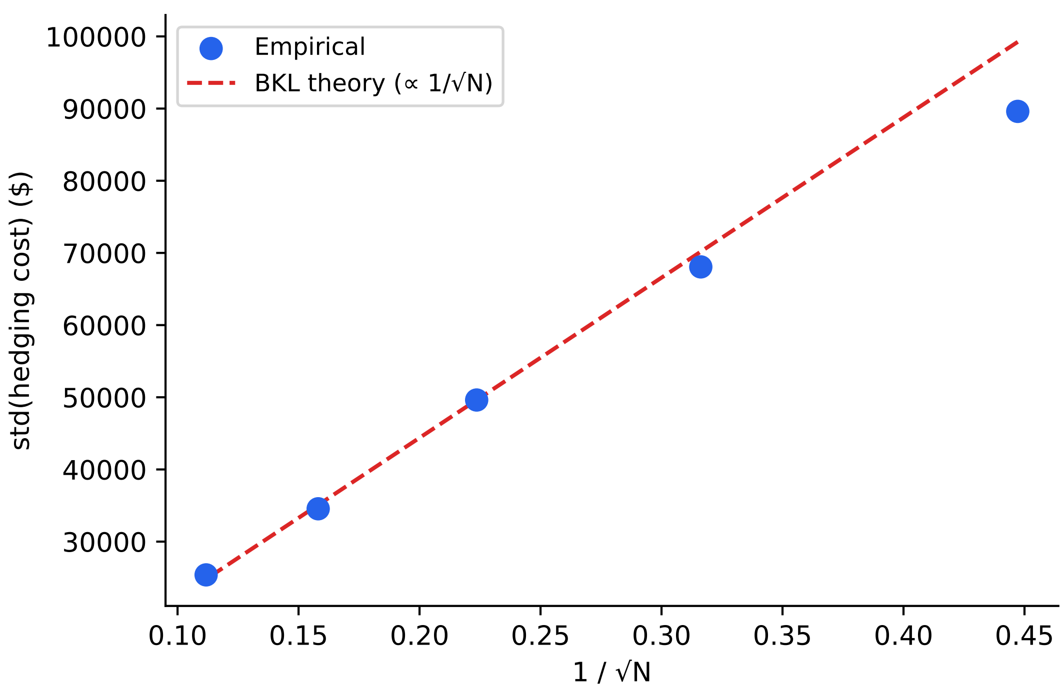

Figure 6.2: Sample std(hedging cost) vs $1/\sqrt{N}$ for N {5, 10, 20, 40, 80} rebalancing steps. The BKL theoretical line is anchored at N=20. Source: synthetic GBM, seed 20260519.

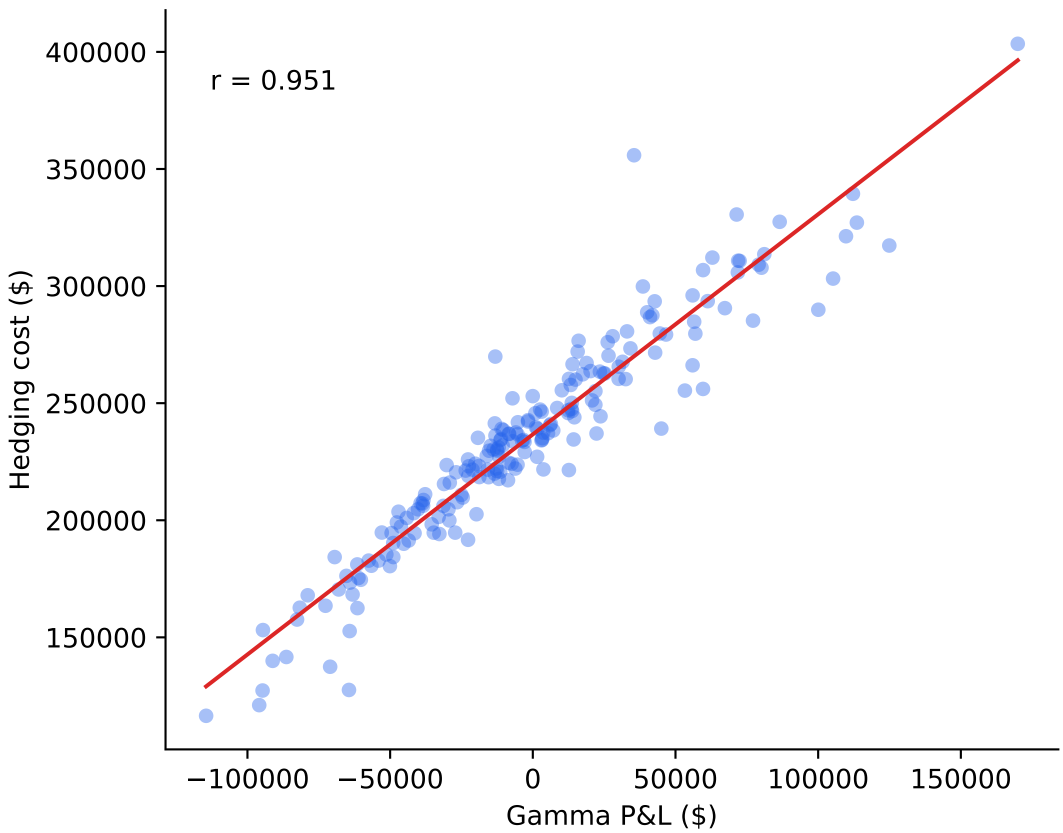

Figure 6.3: Scatter of total hedging cost vs gamma P&L across 200 seeded GBM paths. OLS line and Pearson correlation annotated. Source: synthetic GBM, seed 20260519.

6.3 Carr-Madan gamma attribution

Hedging cost correlates with gamma P&L at r = 0.951 across 200 paths, consistent with the Carr-Madan decomposition Carr and Madan (1998) (Figure 6.3).

6.4 Hull Table 19.2 / 19.3 regression

For Hull Table 19.2 our hedging cost is $253,731 vs the reference $263,300. For Table 19.3: $251,822 vs $256,600. All step certificates pass for both paths (Figure 6.4).

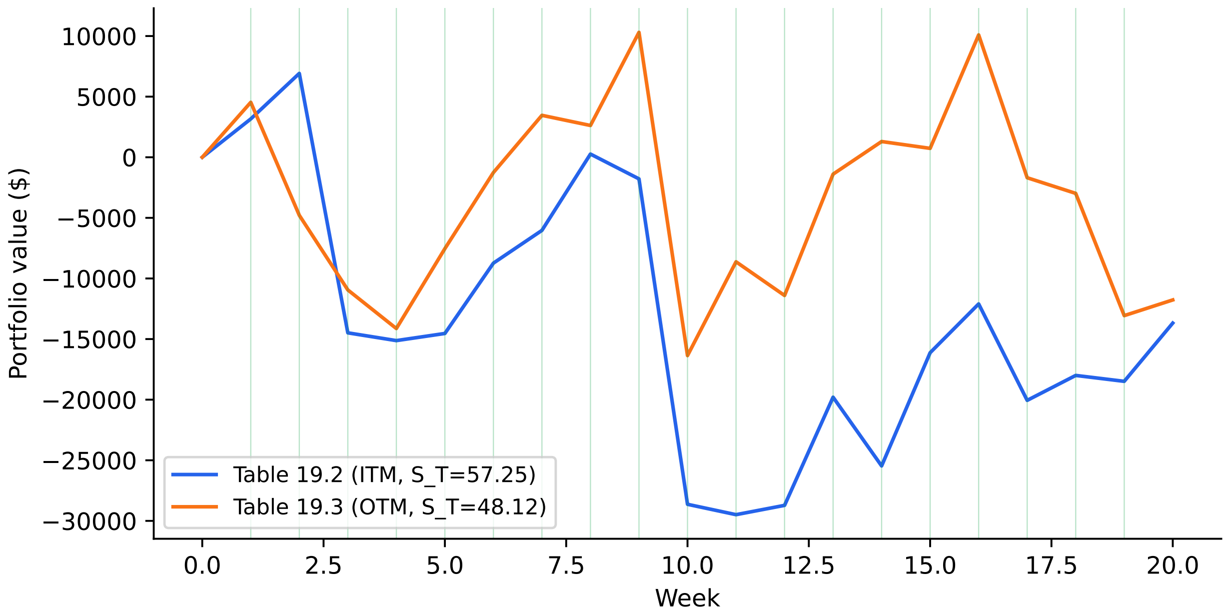

Figure 6.4: Portfolio value path for Hull Table 19.2 (ITM expiry, blue) and Table 19.3 (OTM expiry, orange). Step-certificate pass indicator shown at the bottom (all pass). Source: Hull (2014) Tables 19.2–19.3.

6.5 QuantLib A-B comparison

> **ℹ Note**
>
> TODO (§7 QuantLib A-B prose): Report ql_max_dev and ql_within_1bp from the chunk above. If ql_available is False, note that QuantLib is not installed and defer to the replication instructions in the appendix.

TODO: Draft prose after running with QuantLib installed (`uv sync --group research`).

6.6 Multi-leg portfolios

> **ℹ Note**
>
> TODO (§7 multi-leg prose): Three figures demonstrating portfolio-level delta hedging. Key point: the Lean accounting module accounts for each leg independently via settlement_value_formula, and the portfolio-level invariants (net delta, step certificates) hold across all legs simultaneously. No new Lean theorems required — settlement_value_formula already covers each leg independently.
> Figures: - fig-straddle-settlement: asymmetric settlement on Hull 19.2 (call ITM, put OTM) and Hull 19.3 (call OTM, put ITM) — demonstrates both settlement branches on the same

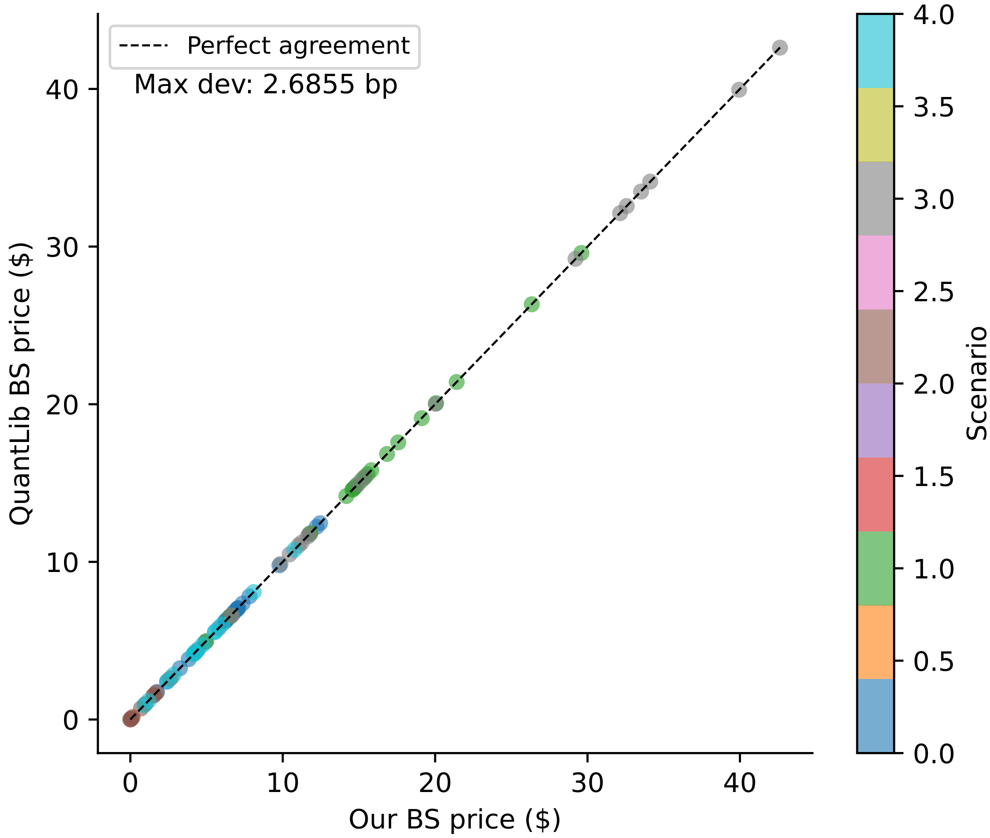

Figure 6.5: Scatter of our BS call price vs QuantLib's analytic price across 5 scenarios $\times$ 20+ evaluation points. Dashed line: 1 bp agreement band. Source: synthetic GBM, seed 20260519.

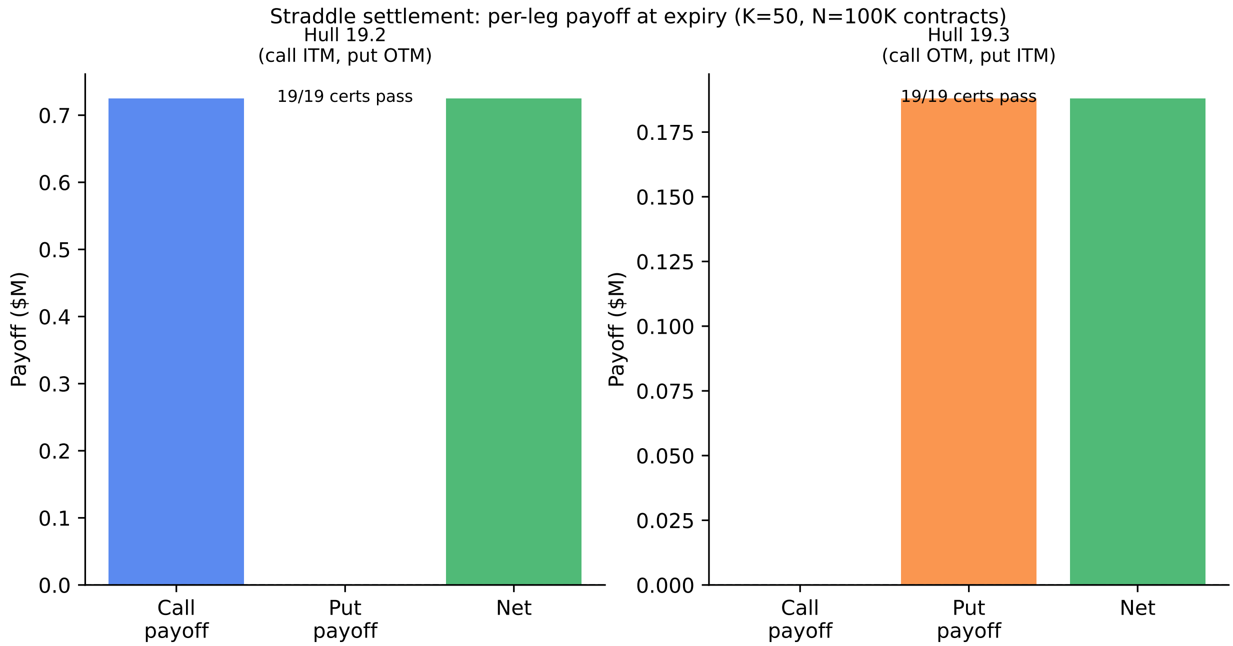

Figure 6.6: Short straddle (short call + short put, K=50) settlement on Hull 19.2 (call ITM, put OTM, S_T=57.25) and Hull 19.3 (call OTM, put ITM, S_T=48.12). Bars show payoff transferred per leg at expiry; all step certificates pass. Source: Hull (2014) Tables 19.2–19.3.

> strategy. - fig-straddle-pnl: cost distribution vs buy-and-hold, 100 GBM paths. - fig-bull-spread: bull call spread accounting path on Hull 19.2 (both legs ITM).

6.6.1 Short straddle: settlement decomposition

6.6.2 Short straddle vs. buy-and-hold

6.6.3 Bull call spread

> **Note**
>
> TODO (§7 multi-leg prose): Three portfolios, each using `run_portfolio_hedge` with `PortfolioStrategy`. The Lean accounting module accounts for each leg via `settlement_value_formula` independently. The net delta hedge is the sum of per-leg deltas at each step. Key results to cite: - Straddle on 19.2: call pays out, put expires worthless — all certs pass. - Straddle on 19.3: put pays out, call expires worthless — all certs pass. - Bull spread on 19.2: both legs ITM, net payoff = $400K — all certs pass.

TODO: Draft prose after verifying all three figures render.

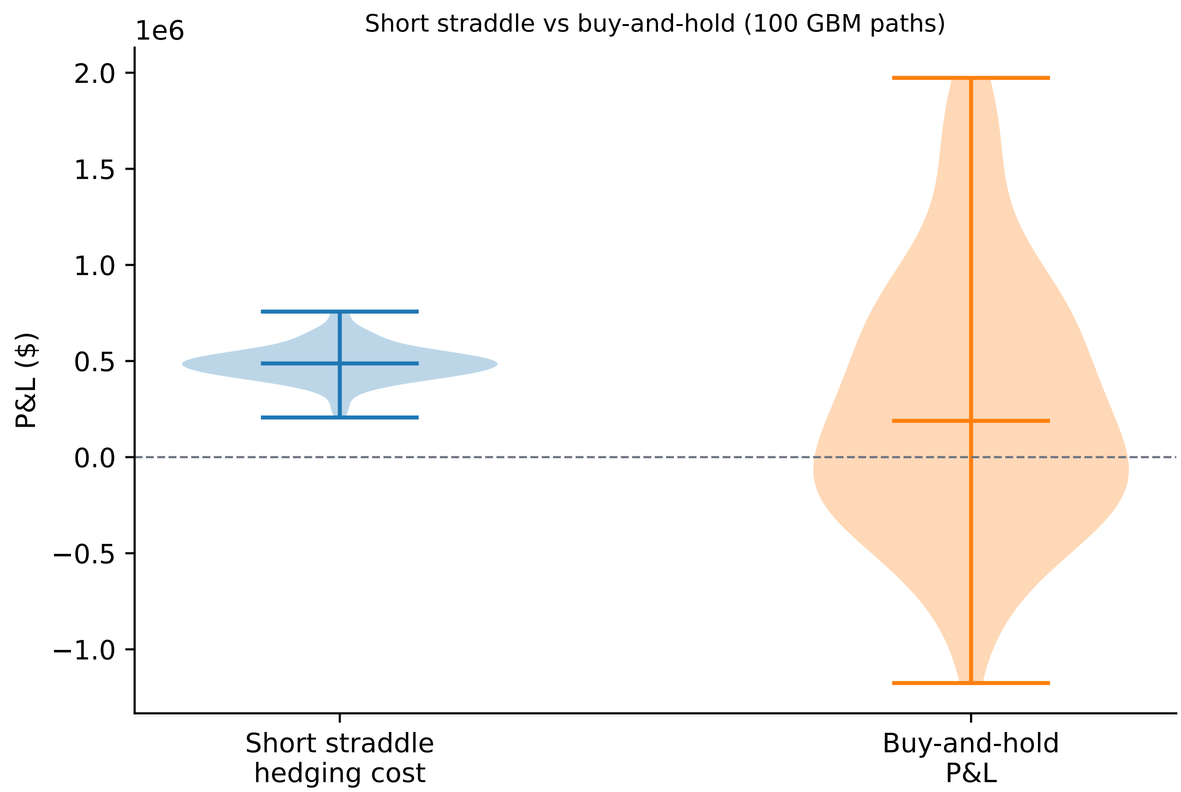

Figure 6.7: Short straddle hedging cost distribution vs. buy-and-hold P&L across 100 seeded GBM paths (S =49, K=50, =20%, T=20 wks). The straddle collects premium from both legs at inception; the hedging cost distribution is wider than a single-leg position because both ITM and OTM settlement branches contribute. Source: synthetic GBM, seed 20261519.

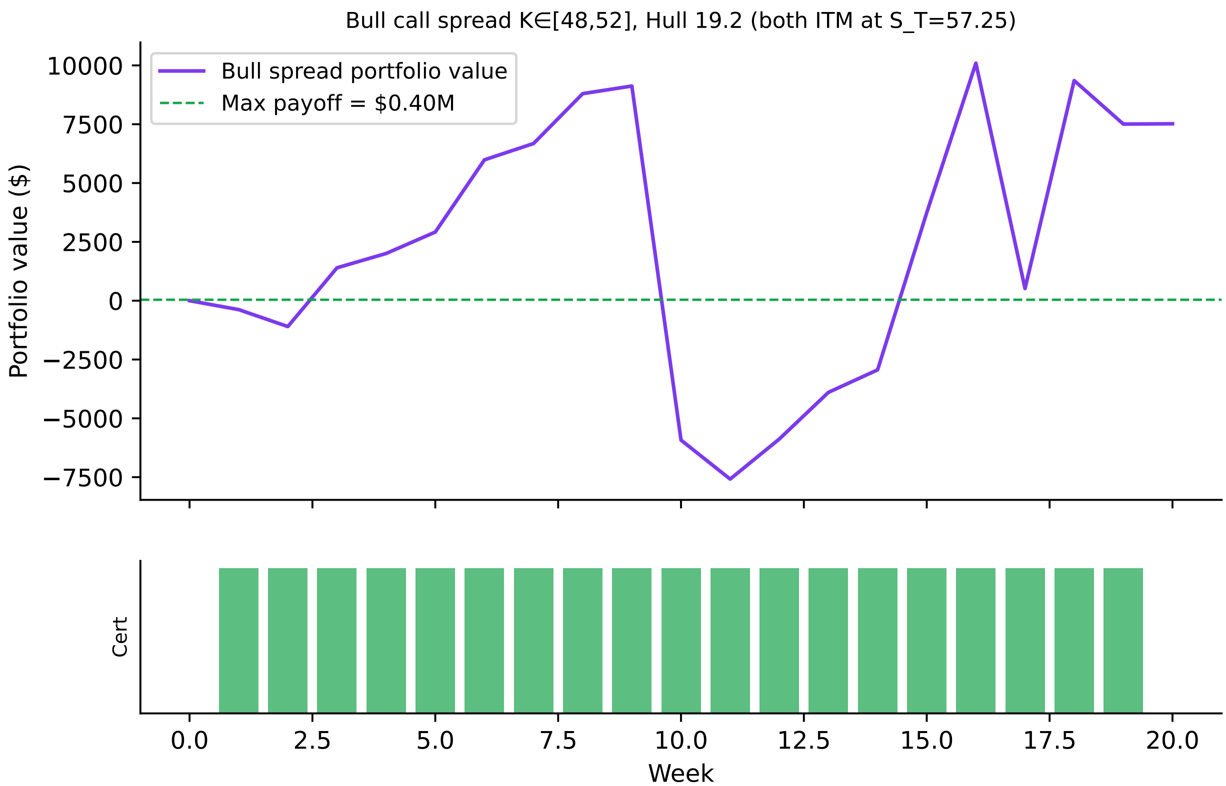

Figure 6.8: Bull call spread (long K=48, short K=52, N=100K) on the Hull 19.2 price path (S_T=57.25). Both calls expire ITM; net payoff = (52−48)×N = \$400K. Portfolio value path (top) and per-step certificate status (bottom, green=pass). Source: Hull (2014) Table 19.2.

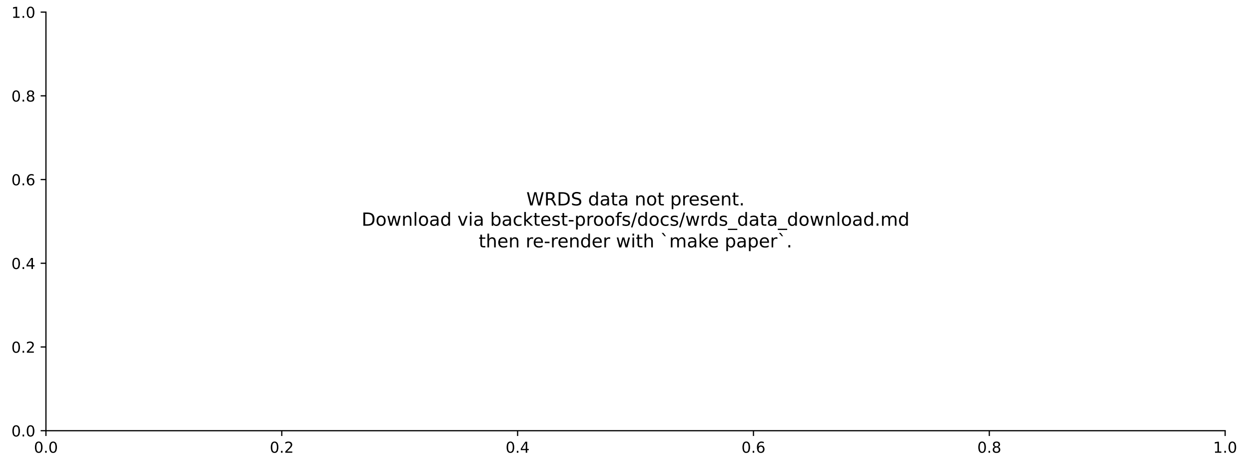

Figure 7.1: Cost/premium distributions across four acute stress windows. Each panel shows the violin of cost/premium ratios for call series that survived data filters. All step certificates pass in every window. Flash Crash 2015 refers to the August 24, 2015 ETF liquidity event, not the May 6, 2010 event. Source: WRDS OptionMetrics (see §5 data table).

6.7 WRDS 2019–2024 holdout

> **Note**
>
> TODO (§7 WRDS holdout prose): Report wrds_holdout_median and wrds_holdout_n. If WRDS_PRESENT is False, note the WRDS download requirement and that stress figures will be added after data arrives.

TODO: Draft after WRDS parquets are downloaded.

7 Robustness

> **Note**
>
> TODO (§8, draft third-ish, alongside §7): Two robustness dimensions required: 1. Sample-period sensitivity: the four stress regimes (COVID, GFC, Volmageddon, Flash Crash) — each is a distinct stressor, not cherry-picking. 2. Specification sensitivity: Leland (1985) rehedge-frequency sweep — does the BKL result hold at other frequencies? 3. Data source A-B: GBM vs WRDS for MC convergence (if WRDS present).

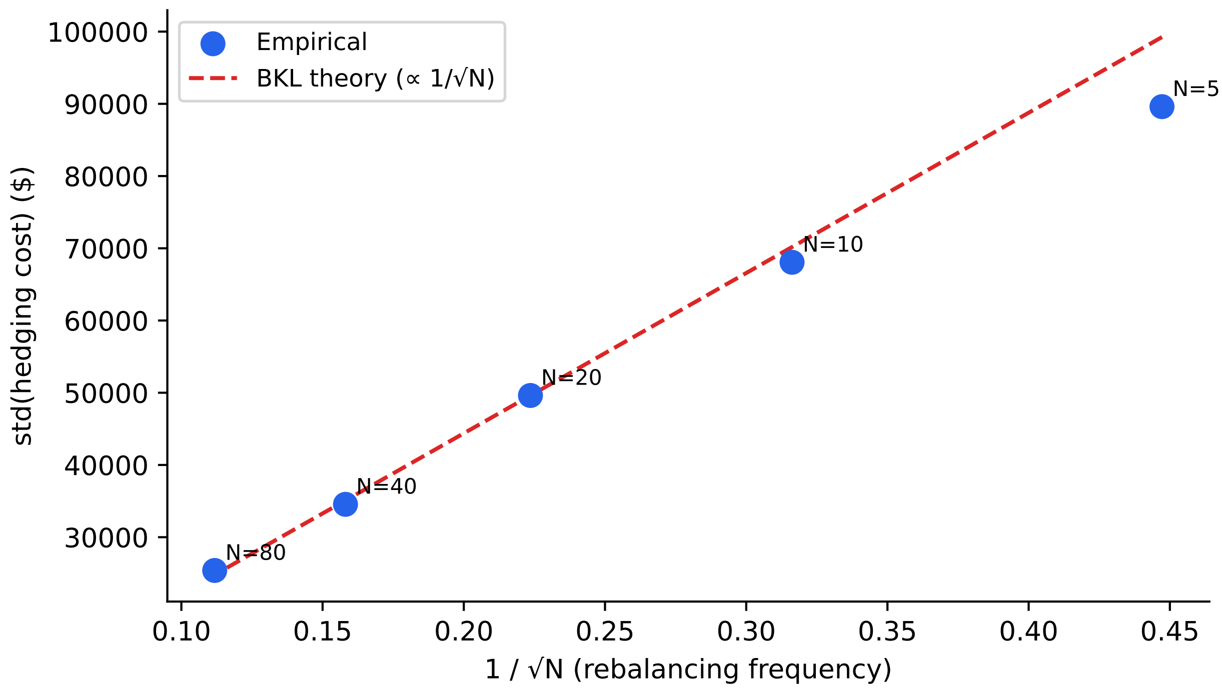

Figure 7.2: Rehedge-frequency variance sweep replicating Leland (1985) Table II. Blue dots: empirical std(cost) for N {5, 10, 20, 40, 80} rebalancing steps (500 paths each). Red dashed: BKL theory (std 1/√N), anchored at N=20. Source: synthetic GBM, seed 20260519.

7.1 Four historical stress regimes

> **Note**
>
> TODO (§8 stress prose): Describe each regime, its stressor, and whether cost/premium is above/below 1.0 and why. State that all certificates pass. Reference the Lean settlement theorem (settlement_value_formula) as the formal guarantee that accounting invariants hold regardless of vol level.

TODO: Draft after WRDS parquets are downloaded.

7.2 Leland (1985) rehedge-frequency sweep

TODO: Draft prose on what the slope ratio says and how it compares to Leland (1985) Table II.

8 Discussion

> **Note**
>
> TODO (§9, draft sixth/last — after all results): 1. What formal proofs guarantee vs empirics: the Lean accounting module proves invariants hold by construction, regardless of the price path. It does NOT prove convergence to BS price (that is an empirical finding, not a theorem). 2. Flat- BS limitation: implied vol varies across the volatility surface; we use inception-date per-series implied vol as (no look-ahead). This understates variance risk during regime changes. 3. Extensions: Carr-Wu (2009) realized-vs-implied vol direction test; mortgage-proofs analogue (multi-agent routing invariants); Leland transaction-cost modification.

TODO: Draft last.

9 Appendix

9.1 A. Replication

```
git clone https://github.com/eigenq-xyz/quant-proofs
cd quant-proofs/backtest-proofs

# 1. Verify Lean proofs (Levels 1-2)
make verify-lean

# 2. Run Python tests (Levels 2-4)
make test

# 3. Install research dependencies and generate figures
cd python && uv sync --group research
cd ..
make paper
```

After downloading WRDS data per `docs/wrds_data_download.md`, re-run `make paper` to regenerate Figure 7.1 with real market data.

9.2 B. Metrics snapshot

References

Carr, Peter, and Dilip Madan, 1998, Towards a theory of volatility trading, *Volatility: New estimation techniques for pricing derivatives*, 417–427.

Echenim, Mnacho, Hervé Guiol, and Nicolas Peltier, 2021, Formalizing the Cox–Ross–Rubinstein pricing of European derivatives in Isabelle/HOL, *Journal of Automated Reasoning* 65, 669–714.

Hull, John C., 2014, *Options, Futures, and Other Derivatives*. 9th Global. (Pearson).